Fundamentals Exam Review

Dosage calculations (opening; revisited 1:25–1:28)

1. 45-lb child prescribed amoxicillin 30 mg/kg/day: daily dose to tenth?

Answer: 613.6 mg/day (45 ÷ 2.2 × 30)

2. Vancomycin 250 mL over 2 hours, 10 gtt/mL: drops/min?

Answer: 21 gtt/min (250×10÷120, rounded

3. Furosemide 40 mg ordered; 4 mg/mL available: volume?

Answer: 10 mL (40÷4

4. D5W 500 mL over 9 hours: pump rate?

Answer: 55.6 mL/hr, or 56 mL/hr if whole-number pump setting required

Medication label — fluconazole (source transcript page 1)

Visible label: Fluconazole for Oral Suspension USP; 40 mg/mL; each teaspoon (5 mL) contains 200 mg fluconazole USP when reconstituted; reconstitute using 24 mL of diluent; 35 mL when reconstituted. Amy confirmed concentration 40 mg/mL. The 24 mL diluent amount is supported by the label, but not by an explicit captured verbal confirmation.



5. What is the concentration on the fluconazole medication label?

Answer: 40 mg/mL.

6. How much diluent is added to reconstitute fluconazole suspension?

Answer: 24 mL

Electrolytes (~8:20–29:00)

7. How quickly should sodium abnormalities be corrected, and why?

Answer: Slowly; rapid correction, especially lowering hypernatremia, risks cerebral edema due to fluid shifts.

8. What system is most associated with sodium imbalance?

Answer: Neurologic ("N for sodium, N for neuro").

9. What fluid state commonly accompanies hyponatremia?

Answer: Hypervolemia / dilution of sodium by excess water (as taught in this review).

10. What fluid state commonly accompanies hypernatremia?

Answer: Hypovolemia / water deficit with concentrated sodium (as taught in this review).

11. What organ is the major concern with potassium disturbances?

Answer: Heart; dysrhythmias.

12. What ECG changes are expected with hyperkalemia?

Answer: Peaked T waves and sometimes widened QRS.

13. Who is at particular risk for hyperkalemia and why?

Answer: End-stage renal disease; impaired urinary potassium excretion.

14. What treatment is given first in Amy's hyperkalemia protocol, and why?

Answer: Calcium gluconate to stabilize the cardiac membrane; dextrose plus insulin shifts potassium intracellularly, and Kayexalate binds potassium in the gut for excretion.

15. What ECG changes are expected with hypokalemia?

Answer: Inverted T waves; possible ST depression.

16. What causes hypokalemia?

Answer: Loop diuretics such as furosemide; malnutrition.

17. How is IV potassium administered?

Answer: Replace potassium using a pump, appropriately diluted, never IV push; Amy mentions 10–20 mEq/hour as an upper-rate range.

18. What electrolyte deficit is especially associated with alcoholism?

Answer: Hypomagnesemia.

19. What is calcium's key association?

Answer: Bones and muscles.

20. What signs suggest hypocalcemia?

Answer: Tingling, numbness, hyperreflexia, tetany, spasms, positive Chvostek and Trousseau signs.

21. What are manifestations/causes of hypercalcemia?

Answer: “Bones, stones, groans”; kidney stones, constipation, bone resorption; elevated PTH, prolonged immobility and potentially vitamin D toxicity.

22. Which vitamin supports calcium absorption?

Answer: Vitamin D.

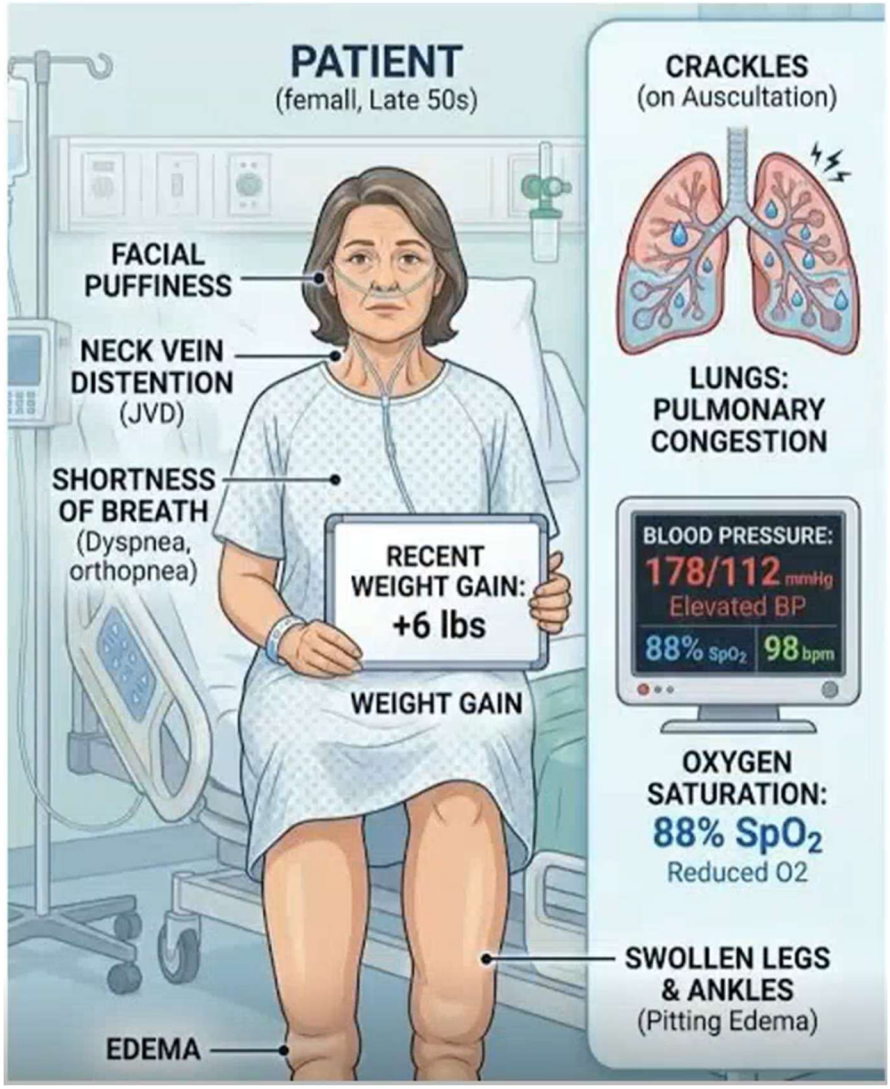
23. What should be monitored when administering magnesium?

Answer: Respiratory depression, especially with rapid administration; Amy says administer slowly.

24. What are manifestations of hypermagnesemia and its treatment?

Answer: Decreased deep-tendon reflexes, respiratory depression, flushing; Amy discusses calcium to stabilize the cardiac membrane.

Fluid volume (~29:45–33:00)



25. What is shown in the first patient picture?

Answer: hypervolemia / fluid-volume excess.

Patient 1 — image transcription: Female, late 50s; facial puffiness; neck vein distention (JVD); shortness of breath (dyspnea, orthopnea); recent weight gain +6 lb; crackles on auscultation; pulmonary congestion; blood pressure 178/112 mmHg; SpO_2 88%; heart rate 98 bpm; swollen legs and ankles with pitting edema.

26. How is hypervolemia managed?

Answer: Diuretics, fluid restriction, daily weights, intake/output, oral hygiene, small cups, gum/hard candy; count ice and gelatin as fluids.



27. What is shown in the second patient picture?

Answer: Amy confirms hypovolemia

Patient 2 — image transcription: Sunken eyes; dry, cracked mucous membranes; decreased skin turgor (slow return after pinching); orthostatic hypotension (dizziness on standing); increased heart rate (tachycardia); oliguria; hypotension; thirst; weakness and fatigue; cool, dry skin; poor skin turgor.

28. How is hypovolemia managed?

Answer: Oral or IV fluids according to severity; daily weights and intake/output; Amy associates it with high sodium and says not to restrict fluids.

Blood transfusion (~33:00–43:20)

29. What must be done before a transfusion?

Answer: Baseline vitals, informed consent, blood type and crossmatch.

30. What is the difference between blood typing and crossmatching?

Answer: Typing identifies blood group; crossmatching tests donor/recipient compatibility.

31. What safety checks and timing apply to routine blood transfusion?

Answer: Two-RN verification; begin within about 30 minutes of obtaining blood; finish within 4 hours; use normal saline in line.

32. How are patients monitored during a transfusion?

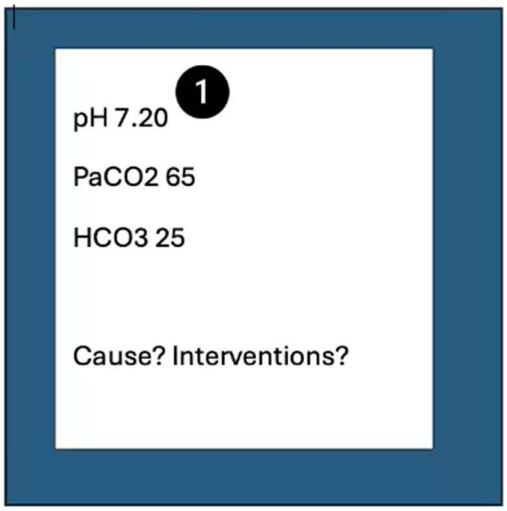
Answer: Stay with patient first 15 minutes, obtain 15-minute vitals, then hourly vitals per Amy's discussion.

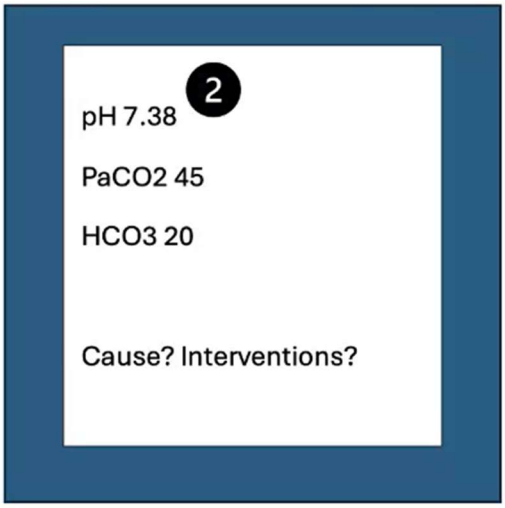
33. What is the priority action if a transfusion reaction is suspected?

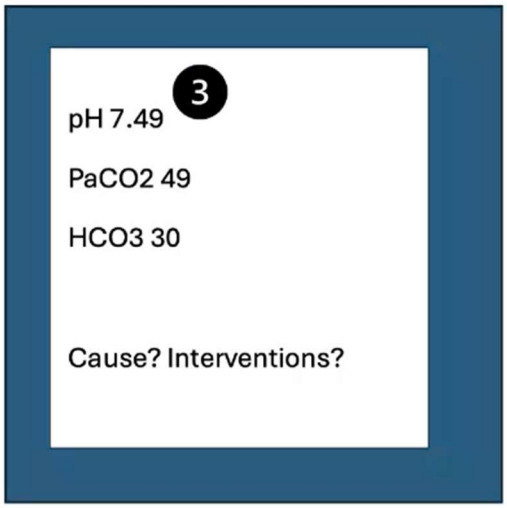
Answer: Stop the transfusion immediately, stay with and assess patient, notify provider, and follow reaction protocol.

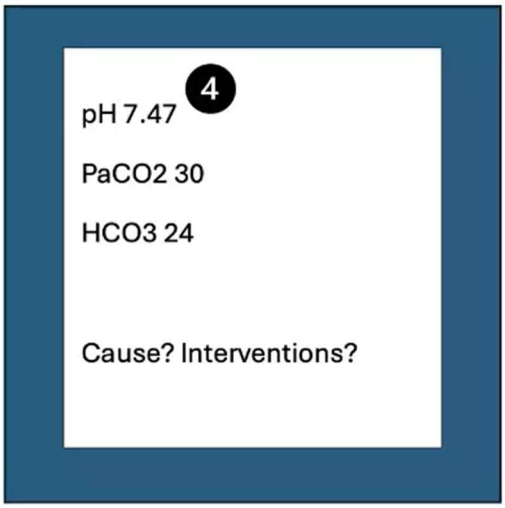
34. What are the major transfusion reactions?

Answer: TACO (circulatory overload), TRALI (acute lung injury), allergic, hemolytic, febrile nonhemolytic.









Original ABG slide — transcribed

Slide	pH	$PaCO_2$	HCO_3^-	Amy's interpretation
1	7.20	65	25	Uncompensated respiratory acidosis
2	7.38	45	20	Fully compensated metabolic acidosis (flagged)
3	7.49	49	30	Partially compensated metabolic alkalosis
4	7.47	30	24	Uncompensated respiratory alkalosis

ABGs (~44:00–59:00)

35. ABG slide #1: what is the interpretation?

Answer: Uncompensated respiratory acidosis: low pH, high CO_2, normal HCO_3^-. Actual numerical values require screenshot.

Screenshot values (ABG #1): pH 7.20; $PaCO_2$ 65 mmHg; HCO_3^- 25 mEq/L.

36. What causes respiratory acidosis, and what interventions did Amy name?

Answer: Hypoventilation/poor gas exchange; COPD, pneumonia, asthma, opioid overdose. Bronchodilator for asthma, pursed-lip breathing/tripod/BiPAP for COPD, naloxone for opioid overdose.

37. ABG slide #2: what is the interpretation?

Answer: Fully compensated metabolic acidosis, per Amy. Actual numerical values require screenshot.

Screenshot values (ABG #2): pH 7.38; $PaCO_2$ 45 mmHg; HCO_3^- 20 mEq/L.

38. What causes metabolic acidosis and how is it managed?

Answer: DKA, lactic acidosis/shock, severe diarrhea, renal failure, starvation/alcohol-related ketoacidosis; treat cause (e.g., IV fluids, insulin, dialysis when appropriate).

39. ABG slide #3: what is the interpretation?

Answer: Partially compensated metabolic alkalosis. Actual numerical values require screenshot.

Screenshot values (ABG #3): pH 7.49; $PaCO_2$ 49 mmHg; HCO_3^- 30 mEq/L.

40. What causes metabolic alkalosis and what symptoms can appear?

Answer: Vomiting, NG suction, loop diuretics; numbness, perioral/finger tingling, spasms. Reassess suction/diuretics and fluids.

41. ABG slide #4: what is the interpretation?

Answer: Uncompensated respiratory alkalosis: high pH, low CO_2, normal HCO_3^-. Actual numerical values require screenshot.

Screenshot values (ABG #4): pH 7.47; $PaCO_2$ 30 mmHg; HCO_3^- 24 mEq/L.

42. What causes respiratory alkalosis?

Answer: Hyperventilation from pain, anxiety, fever, or excessive ventilator settings.

43. What did Amy say about paper-bag rebreathing as an exam answer?

Answer: She says it might be the answer option on their exam, but explicitly warns this is an outdated practice that is generally not used clinically; do not apply routinely to breathless patients.

Safety distinction: Amy described paper-bag rebreathing as a possible test answer while warning it is generally not used clinically and can be unsafe in undifferentiated respiratory distress.

44. How do acid–base states relate to potassium?

Answer: Amy teaches metabolic acidosis with hyperkalemia; metabolic alkalosis with hypokalemia.

Urinary elimination (~59:50–1:12:10)

45. When is an indwelling urinary catheter indicated?

Answer: Amy reviews HOUDINI: gross hematuria, obstruction/retention, urologic surgery, stage 3–4 pressure injury, critical strict I&O, end-of-life care, severe immobility.

46. How do you prevent CAUTI?

Answer: Sterile insertion, catheter care, bag below bladder and off floor, avoid kinks, maintain closed seal, cleanse sampling port, reassess need; empty bag when about half full.

47. What is the first step with suspected urinary retention?

Answer: Assess patient and bladder-scan before straight catheterization; do not jump straight to indwelling Foley.

48. How do you perform a 24-hour urine collection?

Answer: Discard first void and start clock; collect every subsequent void for 24 hours, keep specimen cold as instructed; use another container if needed.

49. How does nephrostomy differ from urinary diversion?

Answer: Nephrostomy drains directly from kidney; urinary diversion reroutes urinary outflow, possibly using intestinal segment/pouch.

50. What are stress, functional and urge incontinence?

Answer: Stress: leakage with coughing/laughing; functional: unable to reach/use toilet due to limitations; urge: sudden urgent need to void.

51. What are polyuria, nocturia, oliguria, anuria, dysuria, hematuria?

Answer: Excess urine; nighttime urination; low urine output; absent urine output; painful/difficult urination; blood in urine, respectively.

Bowel elimination and ostomies (~1:12–1:18)

52. What does a healthy stoma look like, and what is concerning?

Answer: Beefy red, moist, ideally protruding above skin; black/dry or necrotic appearance requires notifying provider.

53. How does ileostomy output compare with colostomy output?

Answer: Ileostomy usually watery; colostomy more formed depending on location; both can produce gas.

54. What complication is especially important with ileostomy?

Answer: Dehydration from watery output; skin breakdown can occur with either ostomy.

55. What does a fecal occult blood test detect?

Answer: Blood in stool.

56. How do fecal impaction and constipation differ?

Answer: Impaction can produce overflow liquid stool around hard stool; urinary retention may accompany impaction.

57. What is correct enema positioning and response to pain?

Answer: Left-side position; if pain occurs, lower enema bag; teach patient to retain fluid as long as tolerated.

Nutrition and infection prevention (~1:18–1:25)

58. Which diet categories did Amy assign to milk, broth, popsicle, apple, apple juice and ice cream?

Answer: Milk and ice cream: full liquid; chicken broth, popsicle, apple juice: clear liquid; apple: regular/solid.

59. What is a mechanical-soft diet example?

Answer: Soft/diced foods such as diced carrots; Amy differentiates this from regular solid foods.

60. How do enteral and parenteral nutrition differ?

Answer: Enteral uses GI tract (NG/G/J tube); parenteral bypasses GI tract through IV access.

61. What precautions apply to enteral feeding?

Answer: Elevate HOB at least 30°, flush before and after, check tube placement and residual as taught; start slowly and titrate as tolerated.

62. How do TPN and PPN differ by access?

Answer: TPN through central line; PPN may run through peripheral line. Maintain sterile line care and scrub hub.

63. What isolation measures for C. difficile?

Answer: Contact precautions with gloves and gown, and handwashing.